1. Variable Separable Method

The simplest and most straightforward technique for solving first-order Ordinary Differential Equations is the **Variable Separable** method. Whenever you are faced with a new DE, you should always check if it can be solved using this method before attempting more complex techniques.

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1. Concept Overview

Definition: A first-order differential equation is considered "separable" if it can be written entirely in the form:

$$f(y)dy = g(x)dx$$

This means you can completely isolate all terms containing the dependent variable y (including dy) on one side of the equals sign, and all terms containing the independent variable x (including dx) on the other side.

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2. Working Rule (Step-by-Step)

To solve a differential equation using the variable separable method, follow these steps: 1. **Rearrange the Equation:** Use algebraic manipulation (factoring, cross-multiplying, dividing) to get all y-terms with dy on the left side, and all x-terms with dx on the right side. 2. **Integrate Both Sides:** Apply the integral sign to both sides of the separated equation: $\int f(y)dy = \int g(x)dx$. 3. **Add the Constant:** After integrating, add a single arbitrary constant of integration (usually C) to one side of the equation (conventionally the x-side). 4. **Apply Initial Conditions (If Given):** If the problem provides specific values for x and y (an Initial Value Problem), plug them into your general solution to solve for the exact value of C.

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3. Solved Questions

Here are four questions from the lecture notes, demonstrating standard separation, factoring, initial value problems, and pre-substitution.

Question 1: Standard Separation

Solve: $(x+1)\frac{dy}{dx} = x(y^2+1)$

Solution: 1. **Separate the variables:** Divide by (y^2+1) and divide by $(x+1)$ to move terms:

$$\frac{1}{y^2+1}dy = \frac{x}{x+1}dx$$

2. **Manipulate the right side for easier integration:** Notice that $\frac{x}{x+1} = \frac{x+1-1}{x+1} = 1 - \frac{1}{x+1}$

$$\frac{1}{y^2+1}dy = \left(1 - \frac{1}{x+1}\right)dx$$

3. **Integrate both sides:**

$$\int \frac{1}{y^2+1}dy = \int \left(1 - \frac{1}{x+1}\right)dx$$

$$\tan^{-1}(y) = x - \ln|x+1| + C$$

4. **Final General Solution:**

$$y = \tan(x - \ln|x+1| + C)$$

Question 2: Factoring Before Separation

Solve: $(xy^2 + x)dx + (yx^2 + y)dy = 0$

Solution: 1. **Factor out common terms:**

$$x(y^2 + 1)dx + y(x^2 + 1)dy = 0$$

2. **Move the dy term to the right side:**

$$x(y^2 + 1)dx = -y(x^2 + 1)dy$$

3. **Separate variables:** Divide by $(x^2 + 1)$ and $(y^2 + 1)$:

$$\frac{x}{x^2 + 1}dx = -\frac{y}{y^2 + 1}dy$$

4. **Integrate both sides:** Multiply both sides by 2 to perfectly match the derivative of the denominators:

$$\frac{1}{2}\int \frac{2x}{x^2 + 1}dx = -\frac{1}{2}\int \frac{2y}{y^2 + 1}dy$$

$$\frac{1}{2}\ln|x^2 + 1| = -\frac{1}{2}\ln|y^2 + 1| + C$$

5. **Simplify using logarithm rules:**

$$\ln(x^2 + 1) + \ln(y^2 + 1) = 2C$$

$$\ln[(x^2 + 1)(y^2 + 1)] = C' \implies (x^2 + 1)(y^2 + 1) = e^{C'} = K$$

(Where K is just a new arbitrary constant).

Question 3: Initial Value Problem (IVP)

Solve: $x\frac{dy}{dx} + \cot y = 0$ given the initial condition $y = \frac{\pi}{4}$ when $x = \sqrt{2}$.

Solution: 1. **Separate the variables:**

$$x\frac{dy}{dx} = -\cot y$$

$$\frac{1}{\cot y}dy = -\frac{1}{x}dx \implies \tan y\,dy = -\frac{1}{x}dx$$

2. **Integrate both sides:**

$$\int \tan y\,dy = -\int \frac{1}{x}dx$$

$$-\ln|\cos y| = -\ln|x| + \ln C$$

(Using $\ln C$ instead of C makes simplification easier). 3. **Simplify the General Solution:** Multiply by -1:

$$\ln|\cos y| = \ln|x| - \ln C \implies \ln(\cos y) = \ln\left(\frac{x}{C}\right)$$

$$\cos y = \frac{x}{C} \implies x = C\cos y$$

4. **Apply Initial Conditions to find C:** Substitute $x = \sqrt{2}$ and $y = \pi/4$:

$$\sqrt{2} = C\cos\left(\frac{\pi}{4}\right)$$

$$\sqrt{2} = C\left(\frac{1}{\sqrt{2}}\right) \implies C = \sqrt{2} \times \sqrt{2} = 2$$

5. **Final Particular Solution:**

$$x = 2\cos y \quad \text{or} \quad y = \cos^{-1}\left(\frac{x}{2}\right)$$

Question 4: Substitution Reducing to Separable

Solve: $x^4 \frac{dy}{dx} + x^3 y = -\sec(xy)$

Solution: 1. **Factor out x^3 on the left side:**

$$x^3 \left(x\frac{dy}{dx} + y \right) = -\sec(xy)$$

2. **Notice the product rule derivative:** The term $(x\frac{dy}{dx} + y)$ is exactly the derivative of the product (xy). Let's use a substitution. * Let $t = xy$ * Differentiate with respect to x: $\frac{dt}{dx} = x\frac{dy}{dx} + y(1) = x\frac{dy}{dx} + y$ 3. **Substitute t and dt/dx into the equation:**

$$x^3 \frac{dt}{dx} = -\sec t$$

4. **Now it is Variable Separable!**

$$\frac{1}{\sec t} dt = -\frac{1}{x^3} dx \implies \cos t \, dt = -x^{-3} dx$$

5. **Integrate:**

$$\int \cos t \, dt = -\int x^{-3} dx$$

$$\sin t = -\left(\frac{x^{-2}}{-2} \right) + C$$

$$\sin t = \frac{1}{2x^2} + C$$

6. **Resubstitute $t = xy$ for the final solution:**

$$\sin(xy) = \frac{1}{2x^2} + C$$

Next Method: What happens if the variables *cannot* be cleanly separated? Proceed to Method 2: Homogeneous Differential Equations to learn the next technique.